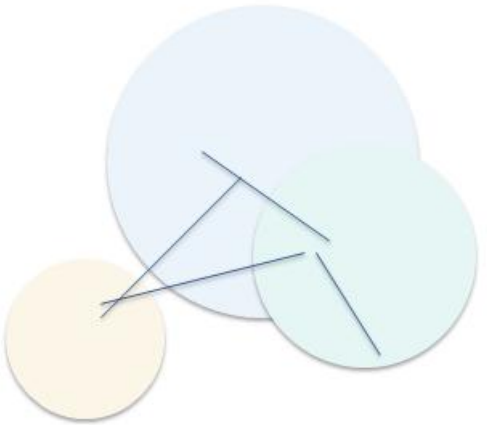
RESEARCH PAPER REVIEW

Specialized Decision Models vs. General-Purpose LLMs

Benchmarking Jev Across Knowledge, Reasoning, and Multilingual Tasks



PRESENTED BY

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A comparative look at specialized decisions and general-purpose language models

General-Purpose LLMs vs. Specialized Decision Models

UNDERSTANDING THE TWO APPROACHES

The study asks whether every structured decision requires a broad, general-purpose language model.

1. General-Purpose LLMs

•Wide range of applications: question answering, text classification, information verification and text generation.
•Flexible capabilities: perform different tasks without being restricted to one application.
•Optimisation: commonly trained using RLHF and RLVF.
•Potential limitation: broad capabilities may be unnecessary for simple, specific decisions.

2. Specialized Decision Models

• Are designed or adapted for particular tasks.
• Can perform better on certain tasks, even when smaller.
•Application-oriented: useful for classification, routing, information verification and other decision-based tasks.

RESEARCH QUESTION

To what extent can specialized decision models match general-purpose LLMs on tasks with fixed answer options, and where do their capabilities begin to differ?

Meet Jev: A Specialized System 1 Decision Model

THE MODEL AT THE CENTRE OF THE STUDY

What is Jev?

- Designed to answer structured questions about an input.
- System 1 approach: makes direct, intuitive decisions rather than deliberate, step-by-step reasoning.
- RLCD (Reinforcement Learning for Calibrated Decisions) trains reliable decisions with confidence scores that reflect accuracy.
- Purpose: focuses on specific decisions instead of general-purpose text generation.

Key capabilities

CHOICE — Selects one answer from up to 255 available choices.

SCORE — Evaluates an input against an ordered scoring rubric.

NOUL — Estimates the probability that a Yes/No statement is true.

Company-reported benefits: 70–500 ms speed • $0.042 per million input tokens • type-safe outputs • confidence scores

Research Objective & Benchmark Datasets

WHAT THE RESEARCHERS SET OUT TO MEASURE

RESEARCH OBJECTIVE

Evaluate Jev against general-purpose LLMs across knowledge, reasoning and multilingual tasks.

8 BENCHMARKS

KNOWLEDGE

Subject knowledge, factual understanding and common sense.

- MMLU
- MMLU-Redux
- MMLU-Pro
- GPQA Diamond
- C-Eval
- HellaSwag
- WinoGrande
- TruthfulQA

3 BENCHMARKS

REASONING

Logic, scientific problem-solving and mathematical reasoning.

- ARC-Challenge
- MathQA
- AQuA-RAT

2 BENCHMARKS

MULTILINGUAL

Answering benchmark questions across languages.

- MMLU-ProX
- MMMLU

13 multiple-choice benchmarks in total: 8 knowledge • 3 reasoning • 2 multilingual

Models & Evaluation Methodology

WHO WAS COMPARED, AND HOW

Jev was compared with 19 general-purpose LLMs across three model tiers.

6 Frontier Models

- GPT-5.6 Sol
- Claude Opus 5.5
- Gemini 3.8 Flash
- Qwen3.8-Flash
- Qwen3.8-Max
- DeepSeek-V4.1-Flash

7 Representative Models

- GPT-4o
- Claude 3.5 Sonnet
- DeepSeek-R1
- DeepSeek-V3
- Qwen3-32B
- Qwen2.5-72B-Instruct
- Llama-3.1-405B-Instruct

6 Small Models

- Qwen3-8B
- Qwen3-4B
- Llama-3.1-8B-Instruct
- Phi-4-mini-instruct
- Gemma-3-4B-IT
- Mistral-7B-v0.3

EVALUATION METHOD

1 Benchmark question + answer options

Jev receives a Choice request.

2 Select an answer

Jev chooses the option it considers most suitable.

3 Compare with key

Check whether the selected answer is correct.

4 Calculate accuracy

Correct answers ÷ total questions × 100%.

Evaluation notes: Jev and frontier models were run once. Representative/small-model scores combine published results and author evaluations. Frontier median is the reference (3 models for GPQA Diamond); differing prompts/settings make some comparisons approximate.

Overall Results: How Does Jev Compare?

A SNAPSHOT OF THE BENCHMARK FINDINGS

The results show a mix of clear strengths and task-specific limitations.

Strong performance

- Best score among evaluated models on MMLU-Redux (95.0%).
- Best score on ARC-Challenge (97.6%).
- Competitive on several knowledge and commonsense benchmarks.

Mathematical limitation

- Fell behind frontier LLMs on MathQA and AQuA-RAT.
- Scored 75.0% on MathQA—17.7 points below the frontier median.
- Scored below all 19 general-purpose LLMs on MathQA.

Multilingual performance

- Scored 83.2% on MMLU-ProX and 85.0% on MMMLU.
- Performed reasonably well on translated questions.
- Had more difficulty with C-Eval's Chinese exam-specific knowledge.

KEY TAKEAWAY

Jev performs strongly on several knowledge and commonsense tasks, but mathematical problem-solving remains a major limitation.

Knowledge vs. Mathematical Performance

WHERE JEV EXCELS—AND WHERE IT STRUGGLES

Knowledge-based tasks

- Top results: best score on MMLU-Redux and ARC-Challenge.
- Competitive: performed well on MMLU, WinoGrande and TruthfulQA.
- Small models: outperformed all small LLMs on non-mathematical comparisons.

Mathematical tasks

- Multi-step problems: struggled with mathematical word problems.
- MathQA: 75.0%, below all 19 general-purpose LLMs.
- AQuA-RAT: scored below all six frontier models.

MATHQA COMPARISON

75.0% Jev

92.7% Frontier median

−17.7 percentage points

POSSIBLE EXPLANATION

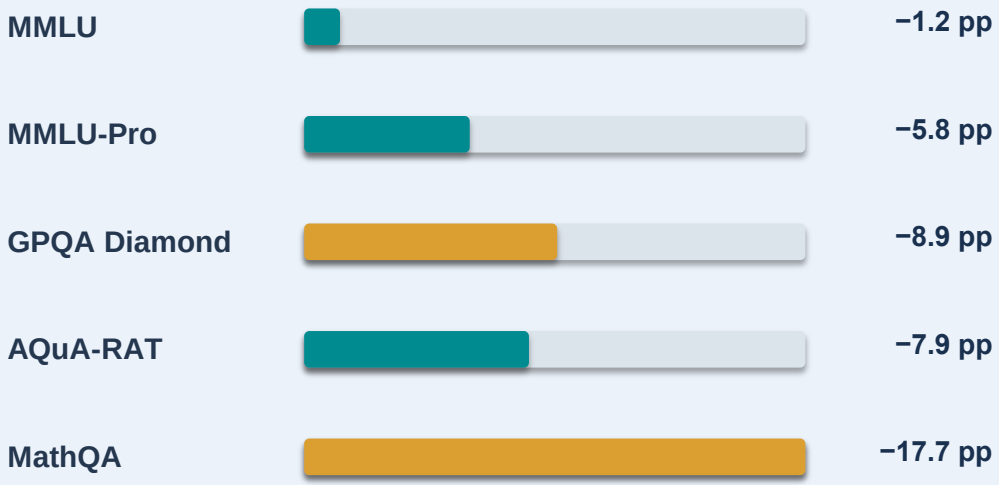
Knowledge questions may involve identifying an option; math word problems require several calculation steps. The authors suggest this may challenge Jev's direct-selection approach.

Reasoning Demands: How Does Task Complexity Affect Jev?

PERFORMANCE DIFFERENCES ACROSS TASKS

The study suggests Jev's performance gap generally increases as tasks demand more complex reasoning.

Jev's gap from the frontier median



Chain-of-Thought (CoT) reasoning

- Prompts an LLM to work through intermediate steps before answering.
- A cited study reported GPT-4o gains of +1.5 points on MMLU and +19.1 on MMLU-Pro.

DeepSeek-R1: Strong representative model

- Strongest representative model; slightly ahead of Jev on average.
- Outperformed Jev on several non-math benchmarks, while Jev did better on the two math benchmarks.

KEY TAKEAWAY

Jev's gap tends to widen on tasks requiring more complex reasoning and calculations.

Multilingual Performance: Can Jev Handle Different Languages?

TRANSLATED QUESTIONS AND NATIVE-LANGUAGE KNOWLEDGE

The study tests whether models can answer benchmark questions presented in different languages.

MMLU-ProX

MMLU-Pro questions translated into 29 languages.

MMMLU

MMLU questions translated into 14 languages.

Jev's multilingual results

	Jev	Frontier median
MMLU-ProX	83.2%	84.5%
MMMLU	85.0%	88.6%

Translated vs. native Chinese questions

- Translated questions: Jev performs reasonably well when English benchmark questions are translated.
- C-Eval: Jev scored 85.3%, around 8.3 points below the frontier median of 93.6%.
- C-Eval uses Chinese exam questions, requiring curriculum-specific knowledge.

KEY TAKEAWAY

Jev transfers reasonably well to translated questions, but performs less strongly on language- and curriculum-specific knowledge.

Reliability Check & Future Directions

HOW CONSISTENT ARE THE FINDINGS, AND WHAT COMES NEXT?

The researchers compared their Jev evaluation with an independent study by Deußer et al. (2026).

Independent evaluation comparison

Benchmark	This paper	Independent	Difference
MMLU	91.3%	91.8%	0.5 pts
C-Eval	85.3%	83.9%	1.4 pts
HellaSwag	89.3%	95.5%	6.2 pts
WinoGrande	91.8%	91.4%	0.4 pts
ARC*	97.6%	98.8%	1.2 pts

**ARC is not like-for-like: this study uses ARC-Challenge; the independent result combines ARC-Easy and ARC-Challenge.*

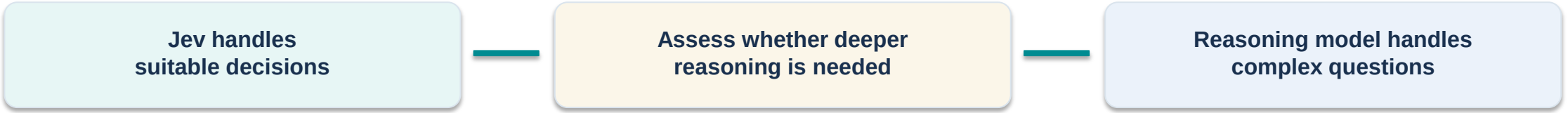
What do the results tell us?

- Close agreement on MMLU, WinoGrande and C-Eval.
- HellaSwag differs by 6.2 points; request templates may affect results because Jev is sensitive to option presentation.

Future directions

- Unified evaluation: test all model tiers using the same protocol.
- Intelligent routing: use Jev's option probabilities to identify questions needing reasoning.

PROPOSED FUTURE WORKFLOW



Proposed future approach

Conclusion: The Right Model for the Right Task

KEY TAKEAWAYS FROM THE RESEARCH

WHAT DID THE RESEARCH FIND?

Jev is competitive on several knowledge and commonsense benchmarks, but struggles with mathematical word problems.

Knowledge

Strong results on MMLU-Redux (95.0%) and ARC-Challenge (97.6%).

Mathematics

Scored below all 19 general-purpose LLMs on MathQA; multi-step calculations remain difficult.

Multilingual tasks

Performs reasonably well on translated questions, but struggles with language- and curriculum-specific knowledge.

Practical use

Effective for structured decisions with fixed options; complex reasoning may benefit from reasoning-focused models.

FINAL RESEARCH INSIGHT

Specialized decision models can be alternatives for constrained tasks, while complex calculations and multi-step reasoning may need general-purpose reasoning models. Model choice should match application requirements.

The best model depends on the task.

Thank You!